

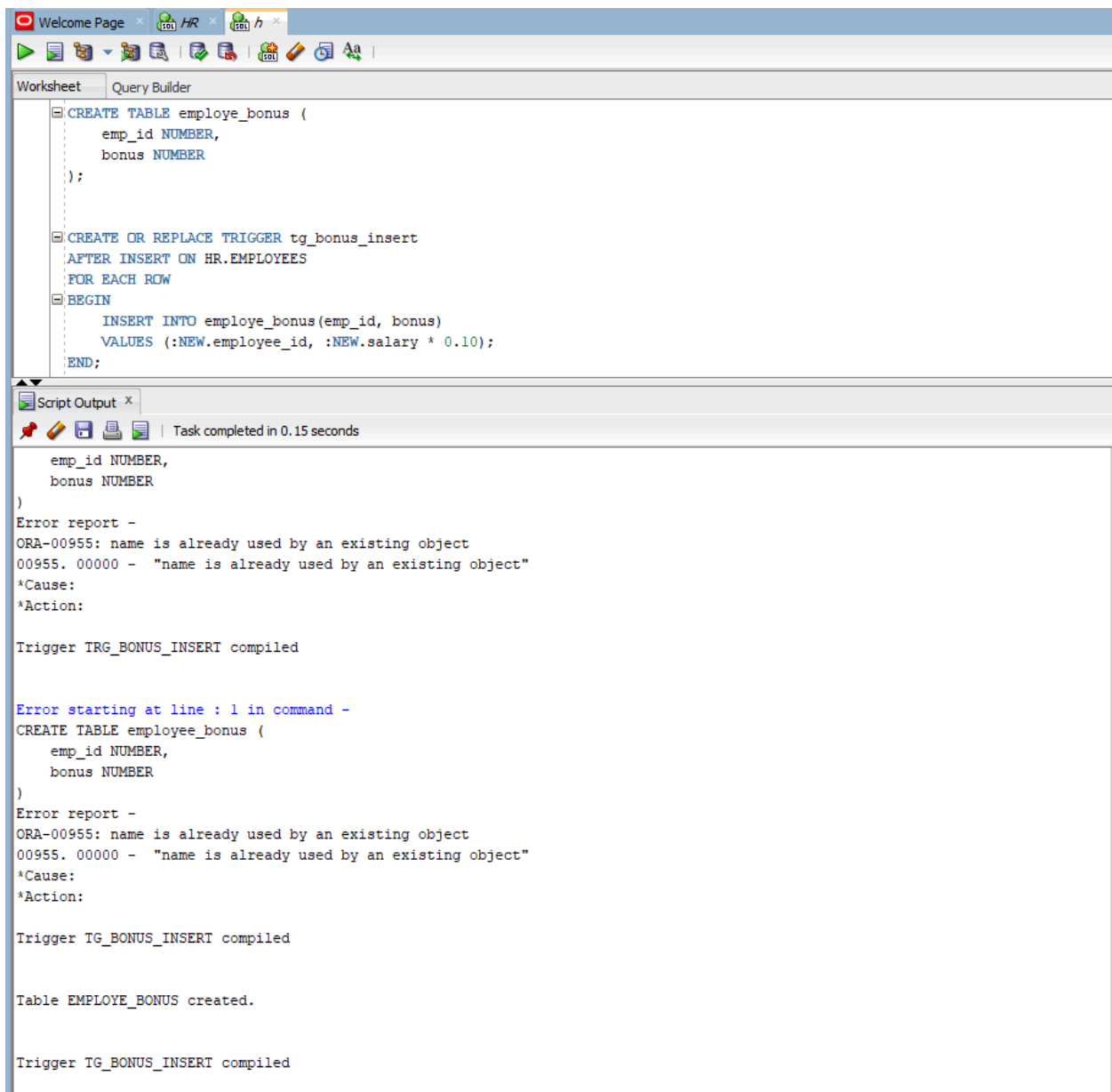
DBS LAB 8

24K-0559

BSCS 4H

DML TRIGGERS:

1).



The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running queries, saving, and other standard database operations. The main window is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Worksheet' pane contains the following SQL script:

```
CREATE TABLE employee_bonus (  
    emp_id NUMBER,  
    bonus NUMBER  
);  
  
CREATE OR REPLACE TRIGGER tg_bonus_insert  
AFTER INSERT ON HR.EMPLOYEES  
FOR EACH ROW  
BEGIN  
    INSERT INTO employee_bonus(emp_id, bonus)  
    VALUES (:NEW.employee_id, :NEW.salary * 0.10);  
END;
```

Below the script, the 'Script Output' pane shows the results of the execution. It indicates that the task was completed in 0.15 seconds. The output includes the table definition, an error report for the first command (ORA-00955: name is already used by an existing object), and the compilation status of the trigger. The second command (CREATE TABLE) also results in an error report (ORA-00955: name is already used by an existing object) and a message stating 'Table EMPLOYEE_BONUS created.'.

emp_id NUMBER,
bonus NUMBER
)
Error report -
ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:
Trigger TRG_BONUS_INSERT compiled

Error starting at line : 1 in command -
CREATE TABLE employee_bonus (
 emp_id NUMBER,
 bonus NUMBER
)
Error report -
ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:
Trigger TG_BONUS_INSERT compiled

Table EMPLOYEE_BONUS created.

Trigger TG_BONUS_INSERT compiled

2).

Oracle SQL Developer : h

The screenshot displays the Oracle SQL Developer interface. On the left, the 'Connections' pane shows a tree of database objects, including 'DEF\$_AQERROR', 'DEF\$_CALLDEST', 'DEF\$_DEFAULTDEST', 'DEF\$_DESTINATION', 'DEF\$_ERROR', 'DEF\$_LOB', 'DEF\$_ORIGIN', 'DEF\$_PROPAGATOR', 'DEF\$_PUSHED_TRANSAC', 'DEPARTMENTS', 'EMPLOYEE_BONUS', 'ENROLLEMENT', 'EVENTS', 'FACULTY', 'FINANCIALRECORD', 'GENRE', 'HELP', 'INVOICE', 'INVOICELINE', 'LOGMNR_AGE_SPILL\$', 'LOGMNR_ATTRCOL\$', 'LOGMNR_ATTRIBUTE\$', and 'LOGMNR_CCOL\$'. The 'Reports' pane on the bottom left lists 'All Reports', 'Analytic View Reports', 'Data Dictionary Reports', 'Data Modeler Reports', 'OLAP Reports', 'TimesTen Reports', and 'User Defined Reports'.

The main workspace is divided into two panes: 'Worksheet' and 'Query Builder'. The 'Worksheet' pane contains the following SQL script:

```
CREATE OR REPLACE TRIGGER trg_salary_check
BEFORE UPDATE OF salary ON HR.employees
FOR EACH ROW
BEGIN
    IF :NEW.salary > 10000 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Salary exceeds limit!');
    END IF;
END;
/
```

The 'Script Output' pane at the bottom shows the execution results:

```
Task completed in 0.074 seconds

Error report -
ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:

Trigger TRG_BONUS_INSERT compiled

Error starting at line : 1 in command -
CREATE TABLE employee_bonus (
    emp_id NUMBER,
    bonus NUMBER
)
Error report -
ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:

Trigger TG_BONUS_INSERT compiled

Table EMPLOYEE_BONUS created.

Trigger TG_BONUS_INSERT compiled

Trigger TRG_SALARY_CHECK compiled
```

3).

Worksheet Query Builder

```
CREATE TABLE deleted_employees_log (  
    emp_id NUMBER,  
    name VARCHAR2(100),  
    salary NUMBER,  
    deleted_at DATE  
);  
  
CREATE OR REPLACE TRIGGER trg_delete_log  
AFTER DELETE ON HR.employees  
FOR EACH ROW  
BEGIN  
    INSERT INTO deleted_employees_log  
    VALUES (:OLD.employee_id, :OLD.first_name, :OLD.salary, SYSDATE);
```

Script Output x

Task completed in 0.138 seconds

Trigger TRG_BONUS_INSERT compiled

Error starting at line : 1 in command -
CREATE TABLE employee_bonus (
 emp_id NUMBER,
 bonus NUMBER
)
Error report -
ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:

Trigger TG_BONUS_INSERT compiled

Table EMPLOYE_BONUS created.

Trigger TG_BONUS_INSERT compiled

Trigger TRG_SALARY_CHECK compiled

Table DELETED_EMPLOYEES_LOG created.

Trigger TRG_DELETE_LOG compiled

4).

Worksheet Query Builder

```
CREATE TABLE salary_log (  
    emp_id NUMBER,  
    old_salary NUMBER,  
    new_salary NUMBER,  
    updated_at DATE  
);  
  
CREATE OR REPLACE TRIGGER trg_salary_log  
AFTER UPDATE OF salary ON HR.employees  
FOR EACH ROW  
BEGIN  
    INSERT INTO salary_log  
    VALUES (:OLD.employee_id, :OLD.salary, :NEW.salary, SYSDATE);
```

Script Output x

Task completed in 0.144 seconds

```
bonus NUMBER  
)  
  
Error report -  
ORA-00955: name is already used by an existing object  
00955. 00000 - "name is already used by an existing object"  
*Cause:  
*Action:  
  
Trigger TG_BONUS_INSERT compiled  
  
Table EMPLOYE_BONUS created.  
  
Trigger TG_BONUS_INSERT compiled  
  
Trigger TRG_SALARY_CHECK compiled  
  
Table DELETED_EMPLOYEES_LOG created.  
  
Trigger TRG_DELETE_LOG compiled  
  
Table SALARY_LOG created.  
  
Trigger TRG_SALARY_LOG compiled
```

DDL TRIGGERS:

1).

Worksheet Query Builder

```

CREATE TABLE audit_log (
    username VARCHAR2(50),
    table_name VARCHAR2(50),
    action_time DATE
);

CREATE OR REPLACE TRIGGER trg_create_table
AFTER CREATE ON DATABASE
BEGIN
    IF ORA_DICT_OBJ_TYPE = 'TABLE' THEN
        INSERT INTO audit_log
        VALUES (USER, ORA_DICT_OBJ_NAME, SYSDATE);
    END IF;
END;
/

```


2).

Oracle SQL Developer interface showing a script in the Query Builder and its execution output in the Script Output window.

Query Builder:

```
CREATE OR REPLACE TRIGGER trg_no_alter_after_hours
BEFORE ALTER ON DATABASE
BEGIN
    IF ORA_DICT_OBJ_NAME = 'EMPLOYEES' THEN
        IF TO_CHAR(SYSDATE, 'HH24') NOT BETWEEN 8 AND 18 THEN
            RAISE_APPLICATION_ERROR(-20002, 'Cannot ALTER after business hours!');
        END IF;
    END IF;
END;
```

Script Output:

Task completed in 0.085 seconds

ORA-00955: name is already used by an existing object
00955. 00000 - "name is already used by an existing object"
*Cause:
*Action:

Trigger TG_BONUS_INSERT compiled

Table EMPLOYE_BONUS created.

Trigger TG_BONUS_INSERT compiled

Trigger TRG_SALARY_CHECK compiled

Table DELETED_EMPLOYEES_LOG created.

Trigger TRG_DELETE_LOG compiled

Table SALARY_LOG created.

Trigger TRG_SALARY_LOG compiled

Trigger TRG_NO_ALTER_AFTER_HOURS compiled

3)

Oracle SQL Developer interface showing the Query Builder window and the Script Output window.

Query Builder Window:

```
CREATE TABLE drop_log (  
    username VARCHAR2(50),  
    object_name VARCHAR2(50),  
    drop_time DATE  
);  
  
CREATE OR REPLACE TRIGGER trg_drop_log  
AFTER DROP ON DATABASE  
BEGIN  
    IF ORA_DICT_OBJ_TYPE = 'TABLE' THEN  
        INSERT INTO drop_log  
        VALUES (USER, ORA_DICT_OBJ_NAME, SYSDATE);  
    END IF;
```

Script Output Window:

Task completed in 0.144 seconds

Table EMPLOYE_BONUS created.

Trigger TG_BONUS_INSERT compiled

Trigger TRG_SALARY_CHECK compiled

Table DELETED_EMPLOYEES_LOG created.

Trigger TRG_DELETE_LOG compiled

Table SALARY_LOG created.

Trigger TRG_SALARY_LOG compiled

Trigger TRG_NO_ALTER_AFTER_HOURS compiled

Table DROP_LOG created.

Trigger TRG_DROP_LOG compiled

4).

Worksheet Query Builder

```
CREATE OR REPLACE TRIGGER trg_prevent_audit_drop
BEFORE DROP ON DATABASE
BEGIN
    IF ORA_DICT_OBJ_NAME = 'AUDIT_LOG' THEN
        RAISE_APPLICATION_ERROR(-20003, 'Cannot drop AUDIT_LOG table!');
    END IF;
END;
```

Script Output x

Task completed in 0.076 seconds

*Cause:
*Action:

Error starting at line : 7 in command -
CREATE OR REPLACE TRIGGER trg_instead_insert
INSTEAD OF INSERT ON dept_view
FOR EACH ROW
BEGIN
 INSERT INTO departments(department_id, department_name)
 VALUES (:NEW.department_id, :NEW.department_name);

 INSERT INTO employees(employee_id, first_name, salary, department_id)
 VALUES (:NEW.employee_id, :NEW.first_name, :NEW.salary, :NEW.department_id);
END;

Error report -
ORA-00942: table or view does not exist
00942. 00000 - "table or view does not exist"
*Cause:
*Action:

Table AUDIT_LOG created.

Trigger TRG_CREATE_TABLE compiled

Trigger TRG_PREVENT_AUDIT_DROP compiled

SYSTEM TRIGGER

1).

Worksheet Query Builder

```
CREATE TABLE system_logs (  
    event_time DATE,  
    status VARCHAR2(50)  
);  
  
CREATE OR REPLACE TRIGGER trg_startup  
AFTER STARTUP ON DATABASE  
BEGIN  
    INSERT INTO system_logs VALUES (SYSDATE, 'DATABASE STARTED');  
END;  
/  
/
```

Script Output x

Task completed in 0.116 seconds

Trigger TRG_SALARY_CHECK compiled

Table DELETED_EMPLOYEES_LOG created.

Trigger TRG_DELETE_LOG compiled

Table SALARY_LOG created.

Trigger TRG_SALARY_LOG compiled

Trigger TRG_NO_ALTER_AFTER_HOURS compiled

Table DROP_LOG created.

Trigger TRG_DROP_LOG compiled

Table SYSTEM_LOGS created.

Trigger TRG_STARTUP compiled

2)

Worksheet Query Builder

```
CREATE TABLE failed_logins (  
    username VARCHAR2(50),  
    attempt_time DATE  
);  
  
CREATE OR REPLACE TRIGGER trg_failed_login  
AFTER SERVERERROR ON DATABASE  
BEGIN  
    IF IS_SERVERERROR(1017) THEN  
        INSERT INTO failed_logins  
        VALUES (SYS_CONTEXT('USERENV', 'SESSION_USER'), SYSDATE);  
    END IF;  
END;
```

Script Output x

Task completed in 0.139 seconds

Trigger TRG_DELETE_LOG compiled

Table SALARY_LOG created.

Trigger TRG_SALARY_LOG compiled

Trigger TRG_NO_ALTER_AFTER_HOURS compiled

Table DROP_LOG created.

Trigger TRG_DROP_LOG compiled

Table SYSTEM_LOGS created.

Trigger TRG_STARTUP compiled

Table FAILED_LOGINS created.

Trigger TRG_FAILED_LOGIN compiled

3)

CREATE TABLE user_activity_log (

username VARCHAR2(50),

login_time DATE,

logout_time DATE,

duration NUMBER

);

CREATE OR REPLACE TRIGGER trg_logout

AFTER LOGOFF ON DATABASE

BEGIN

INSERT INTO user_activity_log

VALUES (

SYS_CONTEXT('USERENV','SESSION_USER'),

NULL,

SYSDATE,

NULL

);

END;

/

INSTEAD OF:

1).

```
worksneet | Query Builder
CREATE OR REPLACE VIEW emp_dept_view AS
SELECT
    e.employee_id,
    e.first_name,
    e.last_name,
    e.salary,
    e.department_id,
    d.department_name
FROM hr.employees e
JOIN hr.departments d
ON e.department_id = d.department_id;

CREATE OR REPLACE TRIGGER trg_emp_dept_insert
INSTEAD OF INSERT ON emp_dept_view
FOR EACH ROW
BEGIN
    BEGIN
        INSERT INTO hr.departments (department_id, department_name)
        VALUES (:NEW.department_id, :NEW.department_name);
    EXCEPTION
        WHEN DUP_VAL_ON_INDEX THEN
            NULL;
    END;

    INSERT INTO hr.employees (
        employee_id,
        first_name,
        last_name,
        salary,
        department_id,
        hire_date,
        job_id
    )
    VALUES (
        :NEW.employee_id,
        :NEW.first_name,
        :NEW.last_name,
        :NEW.salary,
        :NEW.department_id,
        SYSDATE,
        'IT_PROG'
    );
END;
/
```

SQL Developer interface showing a SQL script in the Query Builder tab. The script creates a view and a trigger. The Script Output tab shows the execution results, including an error report and successful compilation messages.

SQL Script:

```
CREATE OR REPLACE VIEW emp_dept_view AS
SELECT
    e.employee_id,
    e.first_name,
    e.last_name,
    e.salary,
    e.department_id,
    d.department_name
FROM hr.employees e
JOIN hr.departments d
ON e.department_id = d.department_id;

CREATE OR REPLACE TRIGGER trg_emp_dept_insert
INSTEAD OF INSERT ON emp_dept_view
FOR EACH ROW
BEGIN
    BEGIN
        INSERT INTO hr.departments (department_id, department_name)
        VALUES (:NEW.department_id, :NEW.department_name);
    EXCEPTION
        WHEN DUP_VAL_ON_INDEX THEN
            NULL;
```

Script Output:

```
END;
Error report -
ORA-00942: table or view does not exist
00942. 00000 - "table or view does not exist"
*Cause:
*Action:

Table AUDIT_LOG created.

Trigger TRG_CREATE_TABLE compiled

Trigger TRG_PREVENT_AUDIT_DROP compiled

View EMP_DEPT_VIEW created.

Trigger TRG_EMP_DEPT_INSERT compiled
```

2).

The screenshot shows the Oracle SQL Developer interface. The top toolbar includes icons for running queries, saving, and other database operations. The 'Query Builder' tab is active, displaying the following SQL code:

```
CREATE OR REPLACE VIEW emp_salary_view AS
SELECT employee_id, first_name, salary
FROM hr.employees;

CREATE OR REPLACE TRIGGER trg_salary_update
INSTEAD OF UPDATE ON emp_salary_view
FOR EACH ROW
BEGIN
    IF :NEW.salary < (:OLD.salary * 0.8) THEN
        RAISE_APPLICATION_ERROR(-20001, 'Salary cannot be reduced by more than 20%');
    END IF;

    UPDATE hr.employees
    SET salary = :NEW.salary
    WHERE employee_id = :OLD.employee_id;
END;
```

Below the code editor, the execution results are displayed in two sections:

- View EMP_SALARY_VIEW created.**
- Trigger TRG_SALARY_UPDATE compiled**